

BOM Alternatives

XBandDigitalModule_20x20 - BOM Alternatives

Document Information

Field	Value
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1. Primary vs Alternative Components

1.1 Critical Components

Component	Primary	Alternative	Trade-off
ADC	ADC12DJ5200-SP	ADC12DJ5200-SEP	Lower TID (30kr
FPGA	XQRVC1902	XQRKU060	Lower speed (12
Clock Synth	LMX2615-SP	LMX2694-SEP	Lower TID, smal
Clock Dist	LMK04832-SP	LMK04832-SEP	Lower TID, plas
Buck Controller	TPS7H5002-SP	RIC70847	Newer, differen
LDO	TPS7H1111-SP	ISL75052SEH	Higher noise, d

1.2 Passive Components

Component	Primary	Alternative	Notes
MLCC 10uF 25V	TDK C3225X5R1E1	Murata GRM31CR6	Equivalent
MLCC 10uF 10V	TDK C3225X5R1A1	Samsung CL32A10	Equivalent
MLCC 100nF 16V	Murata GRM188R7	TDK C1005X7R1C1	Equivalent
Inductor 4uH	Coilcraft XAL60	Würth 744355140	Verify saturati
Ferrite Bead	Murata BLM18AG1	TDK MPZ2012S102	Equivalent

2. Multi-Source Strategy

2.1 Critical Components (Single Source)

Component	Vendor	Reason	Risk
ADC12DJ5200-SP	Texas Instrumen	Only space-grad	High
XQRVC1902	AMD/Xilinx	Only space-grad	High
LMX2615-SP	Texas Instrumen	Only QML synthe	Medium

2.2 Dual-Source Components

Component	Source 1	Source 2	Notes
Buck Regulator	ISL70003ASEH (R	ISL71001SLHM (R	Same vendor, di
LDO	TPS7H1111-SP (T	ISL75052SEH (Re	Different vendo
Sequencer	TPS7H3014-SP (T	ISL70321SEH (Re	Different vendo
MOSFET	CSD19538Q3A (TI	IRFB4110PBF (IR	Different vendo

3. Cost Analysis

3.1 Unit Cost Breakdown

Category	Primary Cost	Alternative Cos	Savings
ADC	\$850	\$650 (SEP)	\$200
FPGA	\$180,000	\$100,000 (KU060)	\$80,000
Clock	\$450	\$350 (SEP)	\$100
Power	\$334	\$300	\$34
Passive	\$100	\$90	\$10
Mechanical	\$2,600	\$2,500	\$100
Assembly	\$5,000	\$5,000	\$0
Total **\$189,334** **\$108,890** **\$80,444**			

3.2 Quantity Price Breaks

Quantity	Unit Price	Total	Notes
1	\$189,334	\$189,334	Single unit
5	\$165,000	\$825,000	Small batch
10	\$145,000	\$1,450,000	Production run
25	\$125,000	\$3,125,000	High volume

3.3 Non-Recurring Engineering (NRE)

Item	Cost	Notes
PCB Fabrication	\$5,000	12-layer, high-
Stencil	\$500	For SMT assembl
Test Fixtures	\$10,000	Custom test jig
Qualification	\$50,000	Radiation, envi
Documentation	\$5,000	Technical manua
Total NRE **\$70,500** -		

4. Availability Assessment

4.1 Current Stock (Estimated)

Component	Distributor Sto	Lead Time
ADC12DJ5200-SP	50 units	16-20 weeks
XQRVC1902	10 units	20-26 weeks
LMX2615-SP	100 units	12-16 weeks
LMK04832-SP	80 units	12-16 weeks
ISL70003ASEH	200 units	8-12 weeks
TPS7H1111-SP	150 units	8-12 weeks

4.2 Risk Items

Component	Risk Level	Mitigation
XQRVC1902	High	Order immediate
ADC12DJ5200-SP	Medium	Order in advanc
LMX2615-SP	Low	Adequate stock

5. Derating Summary

5.1 Voltage Derating

Component	Operating	Rated	Derating
ADC12DJ5200-SP	1.0V	1.1V	91%
XQRVC1902 (VCCI	1.0V	1.05V	95%
MLCC 25V	5.5V	25V	22%
MOSFET 100V	5.5V	100V	5.5%

5.2 Current Derating

Component	Operating	Rated	Derating
ISL70003ASEH	4A	9A	44%
TPS7H1111-SP	1.2A	1.5A	80%
MOSFET CSD19538	44A	100A	44%

5.3 Temperature Derating

Component	Operating	Rated	Derating
FPGA	120 degC	125 degC	96%
ADC	110 degC	125 degC	88%
LDO	100 degC	125 degC	80%

6. Revision History

Version	Date	Author	Description
1.0	2026-09-09	Procurement Eng	Initial release
